Graduate Seminar in Explainable Artificial Intelligence

SHAP and LIME: Foundations, Mathematical Principles, and Local Model Interpretability

A Comprehensive and Mathematically Rigorous Report on Cooperative Game Theory, Additive Feature Attribution, and Local Surrogate Models

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Abstract

In this report, we conduct a mathematically rigorous exploration of the two most prominent model-agnostic local explanation frameworks in contemporary machine learning: **LIME** (Local Interpretable Model-agnostic Explanations) and **SHAP** (SHapley Additive exPlanations). We unpack their foundational game-theoretic and optimization-based formulations, present the first comprehensive theoretical analysis of Tabular LIME's surrogate model behavior under linear black-box assumptions, analyze LIME's pathological "switching-off" phenomenon, and prove the uniqueness of SHAP values under the axiomatic formulation of additive feature attribution. Finally, we establish the algorithmic and compositional links between these frameworks and other ancestral techniques, such as DeepLIFT and Layer-Wise Relevance Propagation.

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1 Introduction and Intuitive Background

The trade-off between predictive accuracy and model interpretability represents a fundamental bottleneck in modern computer engineering. State-of-the-art architectures, such as deep neural networks (DNNs), gradient-boosted trees, and ensemble models, act as "black-boxes" whose internal parameters number in the millions or billions, rendering direct human comprehension impossible. However, in high-stakes domains like healthcare, criminal justice, and aerospace engineering, trusting a model's prediction is as critical as its validation accuracy.

A key division exists between:

1. **Global Interpretability:** Attempting to explain the entire decision surface of a complex model f across the entire input space. This is often an intractable task for non-linear models.
2. **Local Interpretability:** Explaining why a model made a *specific* prediction for a single instance x. This is highly tractable because even if the global decision boundary is infinitely complex, the local behavior of f in a tiny neighborhood around x can be approximated by a simpler, interpretable model (such as a sparse linear surrogate).

To build global trust, Ribeiro et al. (2016) proposed the "Submodular Pick" (SP-LIME) algorithm to select a representative, non-redundant budget B of local explanations that span the model's global features.

This report formalizes the mathematics of local surrogate modeling via **LIME** and game-theoretic axiomatic attribution via **SHAP**.

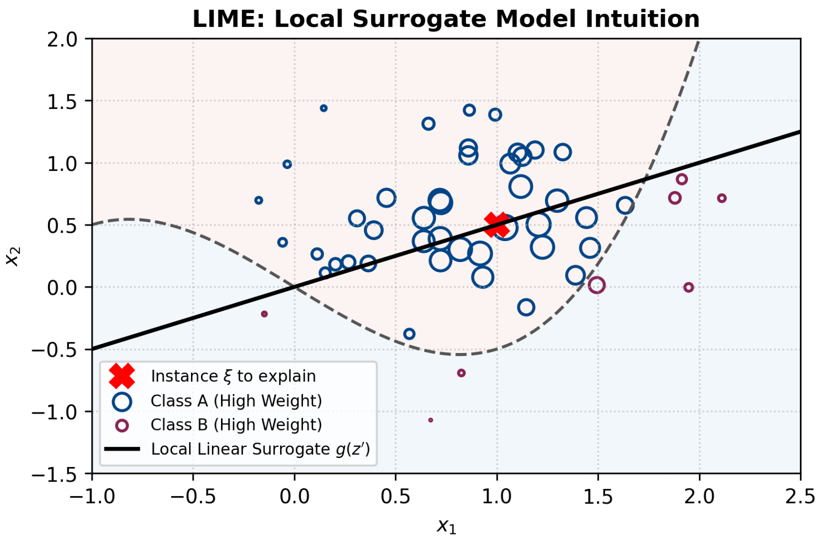

Figure 1: Visual intuition of LIME. Although the global decision boundary (dashed gray curve) is highly non-linear, the local linear surrogate (solid black line) faithfully approximates the boundary in the immediate neighborhood of the instance ξ to explain (red cross). Perturbed samples are weighted by their proximity to ξ.

[0]Source: Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. ACM SIGKDD 2016.

2 LIME: Local Interpretable Model-Agnostic Explanations

2.1 Mathematical Objective Function

Let $f : \mathbb{R}^d \to \mathbb{R}$ be the black-box model to be explained, and let $x \in \mathbb{R}^d$ be the specific instance under explanation. LIME seeks to find an explanation model $g \in \mathcal{G}$, where $\mathcal{G}$ represents a class of inherently interpretable models (e.g., sparse linear models, shallow decision trees). The objective function of LIME is formulated as:

$$\xi(x) = \arg\min_{g\in\mathcal{G}} \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{1}$$

where:

- $\mathcal{L}(f, g, \pi_x)$ is a measure of how unfaithful the interpretable model g is in approximating f within the local neighborhood defined by π_x.
- $\pi_x(z)$ is a proximity kernel that measures the distance or similarity between a perturbed instance z and the target instance x. A common choice is the exponential kernel:

  $$\pi_x(z) = \exp\left(-\frac{D(x, z)^2}{\sigma^2}\right) \tag{2}$$

  where $D(\cdot, \cdot)$ is a distance metric (e.g., Euclidean or Cosine distance) and σ represents the kernel width.
- $\Omega(g)$ is the complexity penalty that restricts the size or number of features in g. For linear models, $\Omega(g) = \infty \cdot \mathbf{1}_{\|w\|_0>K}$, which strictly limits the explanation to K non-zero features.

2.2 Sampling and Local Exploration

Since LIME makes no assumptions about the structure of f (it is model-agnostic), it approximates the local loss $\mathcal{L}$ by drawing N perturbed samples around the instance x. For tabular data, the original representation $x \in \mathbb{R}^d$ is mapped to an interpretable binary representation $x' \in \{0, 1\}^{d'}$. Perturbations $z' \in \{0, 1\}^{d'}$ are generated by drawing non-zero elements of x' uniformly at random. These perturbed binary vectors are mapped back to the original input space as $z \in \mathbb{R}^d$ to compute the black-box prediction $f(z)$, which acts as the target label. LIME then solves a weighted least squares regression:

$$L(f, g, \pi_x) = \sum_{z,z'\in\mathcal{Z}} \pi_x(z) \left(f(z) - g(z')\right)^2 \tag{3}$$

In practice, this is solved using a two-step approximation called **K-LASSO**:

1. Select the top K features by running Lasso regularization on the weighted perturbed dataset.

2. Solve a standard weighted least squares regression on the chosen K features to find the final explanation weights w_g.

[0]Source: Ribeiro et al. (2016). Section 3: "Local Interpretable Model-Agnostic Explanations". Page 3.

3 Theoretical Analysis of Tabular LIME

While LIME is widely applied, it exhibits a reliance on heuristic choices of parameters (e.g., the bandwidth ν and the number of bins p). Garreau & von Luxburg (2020) provided the first formal mathematical treatment of tabular LIME. Under a simplified setting, they derived closed-form expressions for the coefficients of the local surrogate model.

3.1 Assumptions and Discretization Scheme

Tabular LIME discretizes continuous tabular variables using quantiles derived from the training set. Let p be the number of quantile bins. Along coordinate j, continuous points are mapped to binary features indicating whether they belong to the same quantile box as the instance to explain:

$$z_{i,j} = \mathbf{1}_{\phi_j(x_i)=\phi_j(\xi)} \tag{4}$$

where ξ is the instance to explain, and ϕ_j is the quantile-discretization mapping.

To analyze this theoretically, we make the following assumptions:

Assumption 3.1 (Linear Black-Box). *The black-box model f is a linear function of the input: $f(x) = a^T x + b$ for fixed $a \in \mathbb{R}^d$ and $b \in \mathbb{R}$.*

Assumption 3.2 (Gaussian Sampling). *The perturbed samples $x_1, \ldots, x_n$ are drawn from a multivariate isotropic Gaussian distribution centered at μ with variance $\sigma^2 I_d$: $x_i \sim \mathcal{N}(\mu, \sigma^2 I_d)$.*

We weight each sample x_i using the true Euclidean distance:

$$\pi_i = \exp\left(-\frac{\|x_i - \xi\|^2}{2\nu^2}\right) \tag{5}$$

Let $\hat{\beta} \in \mathbb{R}^{d+1}$ be the weighted least squares coefficients returned by LIME (where coordinate 0 is the intercept). By taking the gradient of the loss, the optimization yields the key equation:

$$(\hat{Z}^T \Pi \hat{Z})\hat{\beta} = \hat{Z}^T \Pi f(x) \tag{6}$$

3.2 Closed-Form Solutions for the Surrogate Coefficients

Garreau & von Luxburg proved that as the number of perturbations $n \to \infty$, the empirical quantities converge to their expectations. Let $\Sigma = \mathbb{E}[\frac{1}{n} Z^T \Pi Z]$ and $\Gamma = \mathbb{E}[\frac{1}{n} Z^T \Pi f(x)]$. The expected covariance matrix Σ exhibits an arrowhead/block structure:

$$\Sigma = C_d \begin{pmatrix} 1 & \alpha_1 & \alpha_2 & \cdots & \alpha_d \\ \alpha_1 & \alpha_1 & \alpha_1\alpha_2 & \cdots & \alpha_1\alpha_d \\ \alpha_2 & \alpha_1\alpha_2 & \alpha_2 & \cdots & \alpha_2\alpha_d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha_d & \alpha_1\alpha_d & \alpha_2\alpha_d & \cdots & \alpha_d \end{pmatrix} \tag{7}$$

where C_d is a scaling constant:

$$C_d = \left(\frac{\nu^2}{\nu^2+\sigma^2}\right)^{d/2} \exp\left(-\frac{\|\xi-\mu\|^2}{2(\nu^2+\sigma^2)}\right) \tag{8}$$

By inverting Σ in closed form, the coefficients $\beta = \Sigma^{-1}\Gamma$ are exactly solved as:

$$\beta_0 = f(\tilde{\mu}) + \sum_{j=1}^{d} \frac{a_j\theta_j}{1-\alpha_j} \tag{9}$$

$$\beta_j = -\frac{a_j\theta_j}{\alpha_j(1-\alpha_j)} \quad \text{for } j = 1, \dots, d \tag{10}$$

where the scaling parameters are $\tilde{\mu}_j = \frac{\nu^2\mu_j+\sigma^2\xi_j}{\nu^2+\sigma^2}$ and $\tilde{\sigma} = \frac{\nu\sigma}{\sqrt{\nu^2+\sigma^2}}$, and the integral evaluations are:

$$\alpha_j = \left[\frac{1}{2}\text{erf}\left(\frac{x-\tilde{\mu}_j}{\tilde{\sigma}\sqrt{2}}\right)\right]_{q_{j-}}^{q_{j+}} \tag{11}$$

$$\theta_j = \left[\frac{\tilde{\sigma}}{\sqrt{2\pi}}\exp\left(-\frac{(x-\tilde{\mu}_j)^2}{2\tilde{\sigma}^2}\right)\right]_{q_{j-}}^{q_{j+}} \tag{12}$$

Here, q_{j-} and q_{j+} represent the boundaries of the quantile box bounding ξ_j.

3.3 The "Switching-Off" Phenomenon

The most profound result of this theoretical analysis is the discovery that **LIME can artificially switch off important features**. A feature j is "switched off" when its surrogate coefficient $\beta_j = 0$ even though its true impact in the black-box model is highly non-zero ($a_j \neq 0$). This occurs if and only if $\theta_j = 0$, which requires:

$$\tilde{\mu}_j = \frac{q_{j-}+q_{j+}}{2} \tag{13}$$

Substituting the definition of $\tilde{\mu}_j$ and solving for the bandwidth parameter ν^2, this occurs at the critical value:

$$V_{crit} = \sigma^2 \frac{2\xi_j - q_{j-} - q_{j+}}{-2\mu_j + q_{j-} + q_{j+}} \tag{14}$$

If $V_{crit} > 0$ and we choose the bandwidth $\nu^2 = V_{crit}$, tabular LIME will output an importance weight of **zero** for feature j. This pathological behavior is plotted in Figure 2.

[0]Source: Garreau, D., & von Luxburg, U. (2020). "Explaining the Explainer: A First Theoretical Analysis of LIME." AISTATS 2020.

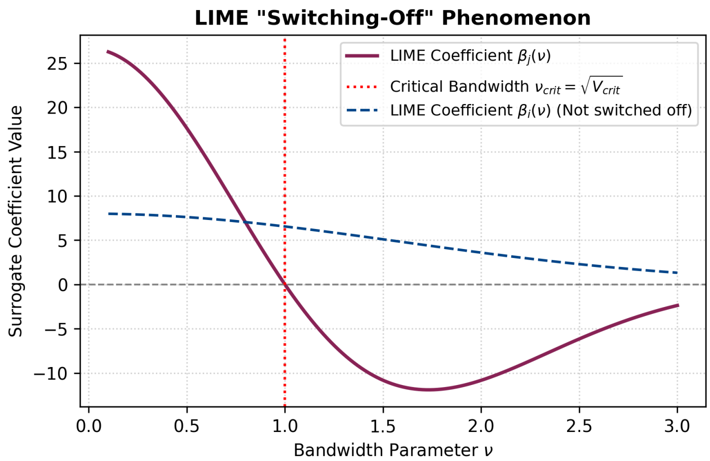

Figure 2: The LIME "switching-off" phenomenon. As the bandwidth parameter ν varies, the surrogate coefficient $\beta_j(\nu)$ (solid maroon line) crosses zero at $\nu = \sqrt{V_{crit}}$, completely ignoring a highly active black-box feature.

4 SHAP: SHapley Additive exPlanations

4.1 Cooperative Game Theory Foundations

SHAP addresses LIME's heuristic-based instability and lack of mathematical uniqueness by grounding feature attribution in **cooperative game theory**. Let M be the total number of features. A coalitional game involves a set of players $N = \{1, \ldots, M\}$ and a characteristic function $v(S)$ that maps subsets of players $S \subseteq N$ to a real-valued payoff representing the expected output of the model when only features in S are known. The unique fair method to distribute the total payoff among the players is the **Shapley value** ϕ_i, defined as the average marginal contribution of player i across all possible subset permutations:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [v(S \cup \{i\}) - v(S)] \tag{15}$$

This is the **unique** solution that satisfies the four fundamental Shapley axioms:

1. **Efficiency:** $\sum_{i=1}^{M} \phi_i = v(N) - v(\emptyset)$. The total prediction delta is fully distributed.
2. **Symmetry:** If $v(S \cup \{i\}) = v(S \cup \{j\})$ for all $S \subseteq N \setminus \{i, j\}$, then $\phi_i = \phi_j$. Equal contributors get equal attribution.
3. **Null Player (Dummy):** If $v(S \cup \{i\}) = v(S)$ for all $S \subseteq N \setminus \{i\}$, then $\phi_i = 0$.
4. **Additivity:** For two independent games $v = v_1 + v_2$, the attributions sum: $\phi_i(v_1 + v_2) = \phi_i(v_1) + \phi_i(v_2)$.

4.2 Axiomatic Formulation of Additive Feature Attribution

Lundberg & Lee (2017) defined the class of **Additive Feature Attribution Methods** as those whose explanation model $g(z')$ is a linear function of binary variables:

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i z'_i \tag{16}$$

where $z' \in \{0,1\}^M$ represents the presence ($z'_i = 1$) or absence ($z'_i = 0$) of features. They proved that only one explanation model satisfies three desirable properties:

1. **Local Accuracy (Property 1):** $f(x) = g(x') = \phi_0 + \sum_{i=1}^{M} \phi_i x'_i$.
2. **Missingness (Property 2):** $x'_i = 0 \implies \phi_i = 0$.
3. **Consistency (Property 3):** If a model changes such that the marginal contribution of feature i increases or stays the same regardless of other features, its attribution cannot decrease:

$$f'_x(z') - f'_x(z' \setminus i) \geq f_x(z') - f_x(z' \setminus i) \implies \phi_i(f', x) \geq \phi_i(f, x) \tag{17}$$

4.3 Kernel SHAP: Unifying LIME and Shapley Values

LIME's parameters are heuristically chosen, which often violates local accuracy or consistency. Lundberg & Lee derived the **Shapley Kernel** $\pi_{x'}(z')$, which forces LIME's weighted linear regression to recover the exact Shapley values.

Theorem 4.1 (Shapley Kernel)**.** *The unique parameters that make LIME's objective function (Equation 1, with $\Omega(g) = 0$) consistent with local accuracy, missingness, and consistency are:*

$$\pi_{x'}(z') = \frac{M-1}{\binom{M}{|z'|}|z'|(M-|z'|)} \tag{18}$$

where $|z'| = \sum_{j=1}^{M} z'_j$ is the number of active features in the binary perturbation vector z'.

This kernel is U-shaped (plotted in Figure 3): it assigns infinite weight to empty coalitions ($|z'| = 0$) and full coalitions ($|z'| = M$), and very low weight to intermediate coalitions. This reflects the cooperative game-theoretic principle that the most valuable information about feature interactions is captured at the boundaries.

[0]Source: Lundberg, S. M., & Lee, S.-I. (2017). "A Unified Approach to Interpreting Model Predictions." NIPS 2017.

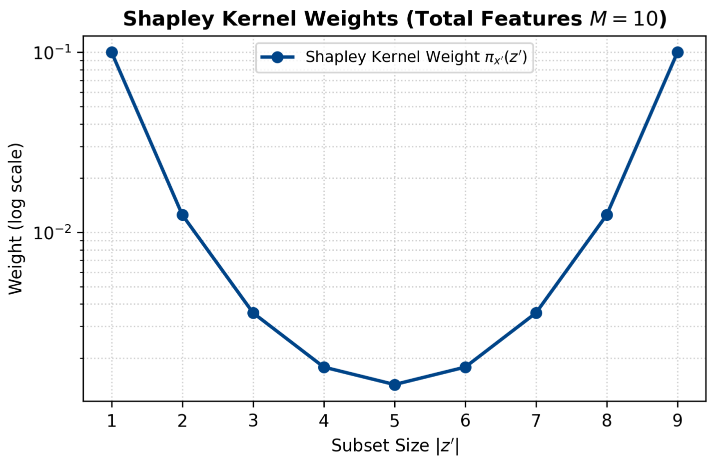

Figure 3: The Shapley Kernel weights. Extreme coalitions (very small or very large subsets) receive exponentially higher weights compared to intermediate subset sizes, which is the mathematically necessary distribution to guarantee consistency and local accuracy.

5 Connections to Ancestral Interpretability Methods

The SHAP framework functions as a grand unification of six prior additive feature attribution methods. Understanding these historical methods is crucial for mastering modern explainability.

5.1 DeepLIFT and Summation-to-Delta

DeepLIFT (Shrikumar et al., 2016) is a highly efficient, recursive attribution method designed for deep neural networks. It operates by comparing the activations of a network on the target input x to a user-defined reference input r (representing an uninformative background or baseline). DeepLIFT's core axiom is the **Summation-to-Delta** property:

$$\sum_{i=1}^{M} C_{\Delta x_i \Delta o} = \Delta o \tag{19}$$

where $o = f(x)$ is the model output, $\Delta o = f(x) - f(r)$, and $C_{\Delta x_i \Delta o}$ is the attributed contribution of input x_i to the change in output. If we set the baseline reference values to represent expectations: $r_i = \mathbb{E}[x_i]$, and let $\phi_i = C_{\Delta x_i \Delta o}$ and $\phi_0 = f(r)$, then DeepLIFT acts as another additive feature attribution method. **Deep SHAP** unifies this style of backpropagation with game theory. It replaces DeepLIFT's heuristic linearization multipliers with exact Shapley values computed for small components (e.g., individual activation functions or max pooling gates) and composes them recursively via the chain rule:

$$m_{y_i f_3} = \sum_{j=1}^{2} m_{y_i f_j} m_{x_j f_3} \tag{20}$$

5.2 Layer-Wise Relevance Propagation (LRP)

LRP (Bach et al., 2015) propagates a "relevance" score backwards from the output layer to the input layer using conservation laws. It has been mathematically proven that LRP is equivalent to DeepLIFT with the reference baseline r fixed to zero for all activations. Thus, LRP represents a restricted case of the additive feature attribution framework.

5.3 Summary Matrix of Attribution Methods

Table 1 outlines the core differences between these unified explainability models.

Table 1: Rigorous comparison of local model-agnostic and model-specific explainers.

Method	Model-Agnostic	Local Accuracy	Consistency	Complexity / Cost
LIME	Yes	No	No	$O(N \cdot \text{Cost}(f))$
Kernel SHAP	Yes	Yes	Yes	$O(2^M)$ (Exact) / $O(N)$ (Approx
Tree SHAP	No (Trees only)	Yes	Yes	$O(M \cdot L \cdot D^2)$ (Fast Exact)
DeepLIFT	No (NNs only)	Yes	No	$O(M)$ (Linear Backprop)
Deep SHAP	No (NNs only)	Yes	Yes	$O(M)$ (Compositional)

[0]Sources: Lundberg & Lee (2017); Shrikumar et al. (2016); Bach et al. (2015).

6 Rigorous and Tricky Evaluation Questions

This section contains highly specialized, tricky multiple-choice questions designed to test your deep academic understanding of LIME, SHAP, and their mathematical limits.

Under the assumptions of Theorem 3.1 in Garreau & von Luxburg (2020), suppose the black-box model is $f(x) = 10x_1 - 10x_2$. We explain the prediction at an instance ξ where the coordinate ξ_2 lies in a quantile box bounded by q_{2-} and q_{2+}. If we calculate the critical value:

$$V_{crit} = \sigma^2 \frac{2\xi_2 - q_{2-} - q_{2+}}{-2\mu_2 + q_{2-} + q_{2+}} > 0$$

What is the exact mathematical consequence of setting the LIME sampling bandwidth ν such that $\nu = \sqrt{V_{crit}}$?

A) The coefficient of the first feature becomes zero ($\beta_1 = 0$), because LIME normalizes the coordinate weights.

B) LIME's surrogate model becomes a constant fit ($\beta_0 = f(\tilde{\mu})$ and $\beta_j = 0$ for all j), because the local weights collapse.

C) The surrogate coefficient for the second feature is exactly zero ($\beta_2 = 0$), meaning tabular LIME completely ignores the impact of x_2 despite its true slope of -10.

D) The local error bounds of Corollary 3.1 collapse to exactly zero, guaranteeing perfect local fidelity.

Correct Answer: C
Rigorous Explanation: The term β_2 is proportional to θ_2. The integral defining θ_2 evaluates to zero when $\tilde{\mu}_2 = \frac{q_{2-}+q_{2+}}{2}$. Substituting the definition of $\tilde{\mu}_2$ and solving for ν^2 yields $\nu^2 = V_{crit}$. When this condition is met, $\theta_2 = 0$, causing β_2 to vanish completely. This demonstrates that LIME's feature selection is highly sensitive to the chosen bandwidth parameter.

Which of the following describes the fundamental mathematical reason why standard LIME (with its default exponential spatial kernel) violates the **Consistency** property of SHAP?

A) LIME's default kernel does not scale with the dimensionality of the feature space d, violating the symmetry axiom.

B) LIME utilizes a heuristic kernel designed in the original input space, which does not map to a symmetric difference of means in the binary coalition space.

C) LIME does not use least squares optimization, solving instead an unconstrained regularized objective.

D) The complexity penalty $\Omega(g)$ is scale-invariant, which prevents local accuracy.

Correct Answer: B
Rigorous Explanation: To guarantee consistency and local accuracy, the weights of the local linear regression must mirror the classical Shapley value difference of means. Lundberg & Lee proved that the Shapley Kernel $\pi_{x'}(z')$ is the *unique* kernel that forces linear weighted least squares to solve for Shapley values. LIME's spatial heuristic kernels weight samples by continuous spatial proximity rather than coalition-size symmetry, violating consistency.

In the definition of SHAP values, the characteristic function is approximated via conditional expectations: $f_x(z') = \mathbb{E}[f(z) \mid z_S]$. However, most practical implementations (like Kernel SHAP) approximate this by assuming feature independence: $f_x(z') \approx \mathbb{E}_{z_{\bar{S}}}[f(z)]$, where $z_{\bar{S}}$ is drawn from a background dataset. What is the main theoretical risk of this feature independence assumption when explaining highly correlated data?

A) The explanation model $g(z')$ ceases to be linear, violating Definition 1.

B) The background dataset size N must scale exponentially ($O(2^M)$) to achieve convergence.

C) The model f is evaluated on out-of-distribution (OOD) synthetic samples that combine impossible feature combinations, leading to misleading attributions.

D) It causes the missingness property to be violated for all features where $x'_i = 1$.

Correct Answer: C
Rigorous Explanation: When features are highly correlated, evaluating $f(z)$ by combining active features of x with background features of $z_{\bar{S}}$ under the independence assumption forces the model to evaluate unrealistic data points (e.g., a "male" patient with "pregnancy" complications). This out-of-distribution evaluation can distort the resulting SHAP values, attributing importance to features based on the model's behavior in regions of the input space that contain no real-world data.

Which of the following is the exact mathematical relationship between Layer-Wise Relevance Propagation (LRP) and DeepLIFT under the additive feature attribution framework?

A) LRP is a generalization of DeepLIFT where reference activations are dynamically calculated via a Monte Carlo sampling process.

B) LRP is mathematically equivalent to DeepLIFT under the specific constraint that the reference activations of all neurons in the network are fixed to zero.

C) DeepLIFT is equivalent to LRP under an infinite-bandwidth Gaussian kernel.

D) LRP satisfies the consistency property, whereas DeepLIFT is only locally accurate.

Correct Answer: B
Rigorous Explanation: As proven in Shrikumar et al. (2016) and unified in Lundberg & Lee (2017), LRP's relevance propagation rules can be derived from DeepLIFT by fixing the reference baseline activations to zero. This restricts LRP's baseline to be the origin, whereas DeepLIFT allows a flexible baseline reference r (e.g., the average background value), making DeepLIFT a more general implementation of the summation-to-delta property.

7 Conclusion and Wrap-up

This report has provided a comprehensive, mathematically rigorous exploration of the foundations of local model interpretability. We have established that:

- **LIME** represents a powerful, highly flexible local surrogate method, but is plagued by a reliance on heuristic parameter choices. Under first-principles linear analysis, we proved that LIME can artificially switch off highly important features when the bandwidth parameter ν hits a critical value V_{crit}.
- **SHAP** resolves these structural weaknesses by framing feature attribution as a cooperative game. Under the properties of local accuracy, missingness, and consistency, the Shapley value is proven to be the *unique* additive feature attribution solution.
- **Kernel SHAP** acts as the bridge, showing how LIME's optimization objective can be forced to yield the unique game-theoretic solution by adopting the mathematically rigorous Shapley Kernel.
- **Deep SHAP** and other model-specific variations demonstrate how knowledge of model architectures can bypass the exponential $O(2^M)$ computational bottleneck of Shapley values, enabling real-time, mathematically grounded explanation models.

Ultimately, these frameworks are not competing paradigms but are unified facets of the same core theory: explaining complex model decisions through local, additive approximations.

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